

Project Workflow

Raw Data



Power Query (Data Cleaning)



Power Pivot (Relationships)



Pivot Tables



Dashboard

Sales Analysis Dashboard

Excel | Power Query | Power Pivot
Interactive Sales Dashboard Using Excel

Sales Analysis Dashboard - 2023

Date (Month)	
Jan	Feb
Mar	Apr
May	Jun
Jul	Aug
Sep	Oct
Nov	Dec

Country	
Algeria	Egypt
Jordan	KSA
Morocco	Qatar
Tunisia	UAE

Category	
Accessories	Electronics
Mobile	Office

Product	
Camera	Charger
Headphones	Keyboard
Laptop	Monitor
Mouse	Printer
Smartphone	Tablet

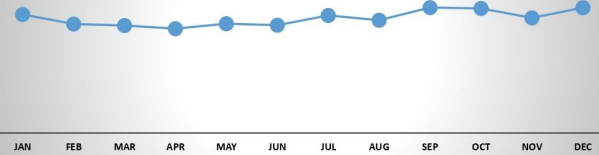
 **Total Sales**
111.2M

 **Total Profit**
22.9M

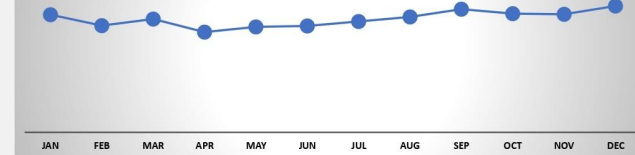
 **Total Orders**
3,000

 **Average Order value**
37.1K

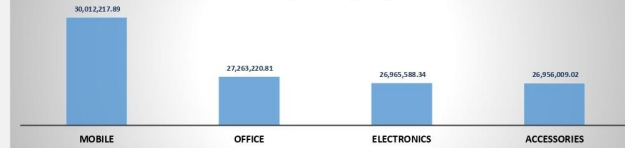
Sales By Month



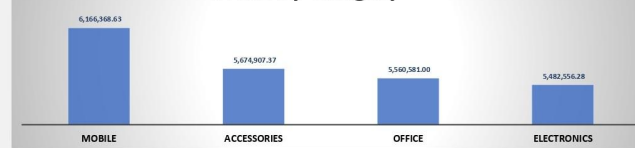
Profit By Month



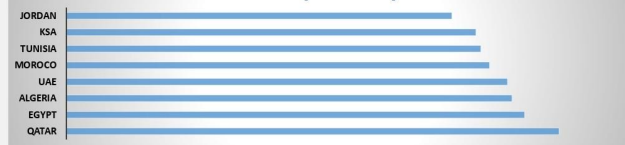
Sales By Category



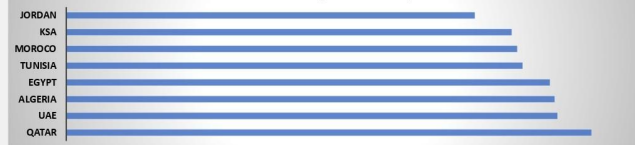
Profit By Category



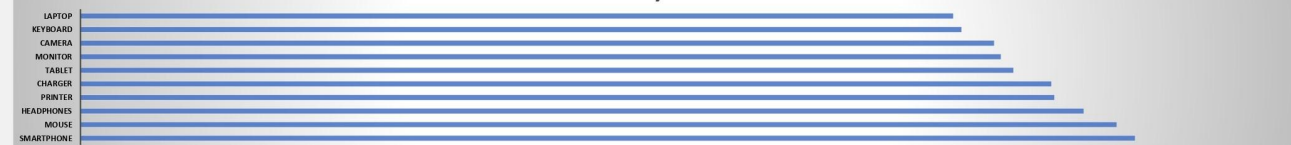
Sales By Country



Profit By Country



Sales By Product



Project Overview

- The objective of this project is to analyze sales performance using Excel and Power Query by cleaning the data, transforming it, and building an interactive dashboard that helps decision-makers monitor sales performance .
- The project focuses on transforming raw sales data into meaningful business insights through an interactive dashboard.

Dataset & Data Cleaning

Data Quality Issues

- ✓ Incorrect product names.
- ✓ Missing country values (28 records).
- ✓ Missing Total Price values.
- ✓ Negative Total Price values (17 records).
- ✓ Missing Cost calculation.

Cleaning Actions

- ✓ Corrected product names using **Replace Values**.
- ✓ Replaced inconsistent Country names in the **Customer** and **Supplier** tables.
- ✓ Filled missing Country values.
- ✓ Calculated missing **Total Price** values.
- ✓ Created a calculated **Cost** column.
- ✓ Identified 17 negative Total Price records, assumed to represent returned orders, and retained them in the dataset.
- ✓ Validated data quality after cleaning.

Query Editor: `= Table.ReplaceValue("#Replaced Value4", null, "Qatar", Replacer.ReplaceValue, {"Country"})`

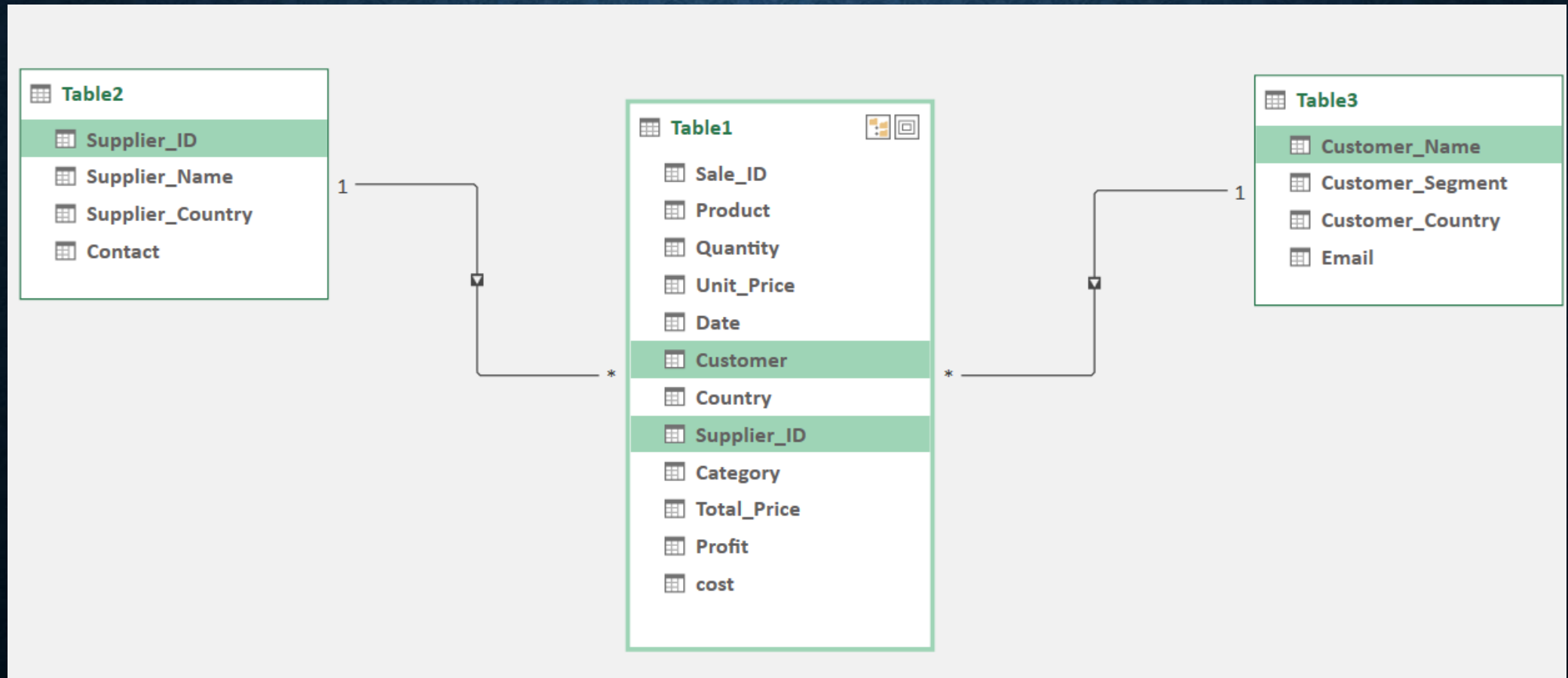
	Sale_ID	Product	Quantity	Unit_Price	Date	Customer	Country
1	24	Camera	34	2729.92	3/9/2023 12:00:00 AM	Customer_81	UAE
2	11	Charger	25	2625.17	7/5/2023 12:00:00 AM	Customer_51	Tunisia
3	28	Laptop	38	1355.67	3/15/2023 12:00:00 AM	Customer_89	UAE
4	15	Smartphone	23	2165.49	11/19/2023 12:00:00 AM	Customer_131	Algeria
5	21	Mouse	24	1860.23	3/1/2023 12:00:00 AM	Customer_178	Tunisia
6	23	Keyboard	26	1189.27	1/27/2023 12:00:00 AM	Customer_187	UAE
7	27	Monitor	11	2708.67	4/19/2023 12:00:00 AM	Customer_40	Jordan
8	18	Charger	25	1088.06	8/1/2023 12:00:00 AM	Customer_1	KSA
9	29	Headphones	14	1665.55	3/7/2023 12:00:00 AM	Customer_125	Jordan
10	17	Mouse	29	772.17	6/20/2023 12:00:00 AM	Customer_177	Jordan
11	14	Tablet	18	1038.89	5/14/2023 12:00:00 AM	Customer_51	Jordan
12	22	Monitor	11	1464.15	3/29/2023 12:00:00 AM	Customer_14	UAE
13	19	Tablet	17	556.33	1/18/2023 12:00:00 AM	Customer_185	Qatar
14	12	Headphones	6	1520.41	3/3/2023 12:00:00 AM	Customer_164	Qatar
15	26	Smartphone	3	2314.25	2/9/2023 12:00:00 AM	Customer_98	Tunisia
16	16	Laptop	6	882.35	10/24/2023 12:00:00 AM	Customer_14	UAE
17	25	Smartphone	28	116.5	10/5/2023 12:00:00 AM	Customer_3	Qatar
18	2001	Camera	13	619.99	9/4/2023 12:00:00 AM	Customer_91	Tunisia
19	2000	Keyboard	12	548.88	11/26/2023 12:00:00 AM	Customer_47	KSA
20	1999	Charger	18	1348.64	1/28/2023 12:00:00 AM	Customer_84	Qatar
21	1998	Keyboard	29	2859.35	7/2/2023 12:00:00 AM	Customer_147	Morocco
22	2006	Laptop	16	1988.2	9/7/2023 12:00:00 AM	Customer_174	Algeria
23	2007	Camera	48	903.07	11/4/2023 12:00:00 AM	Customer_64	Qatar
24	2008	Smartphone	24	670.94	11/30/2023 12:00:00 AM	Customer_119	KSA
25	2005	Tablet	45	1006.54	5/2/2023 12:00:00 AM	Customer_89	Qatar

Query Settings: Name: Table1

APPLIED STEPS: Source, Navigation, Changed Type, Replaced Value, Replaced Value1, Filtered Rows, Replaced Value2, Added Custom, Added Custom1, Added Custom2, Changed Type2, Removed Columns, Changed Type1, Added Custom3, Added Custom4, Changed Type3, Sorted Rows, Removed Columns1, Reordered Columns, Renamed Columns, Added Custom5, Changed Type4, Removed Columns2

Data Modeling

- Created relationships using Power Pivot
- Connected Sales, Customers, and Suppliers tables
- Built a relational data model



Dashboard

Sales Analysis Dashboard - 2023

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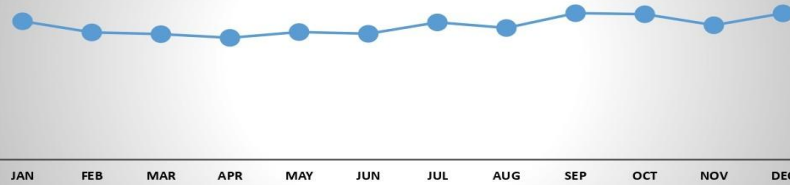
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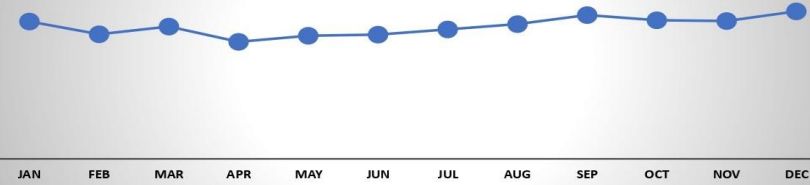
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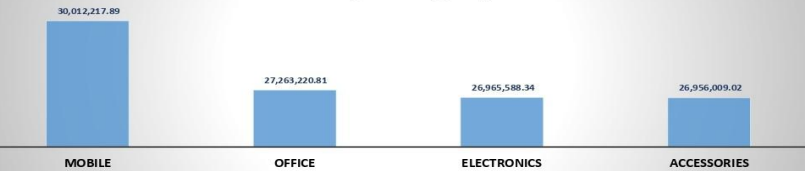
Sales By Month



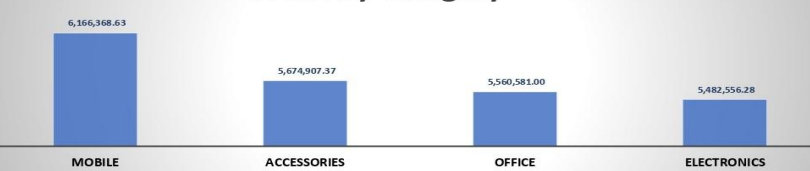
Profit By Month



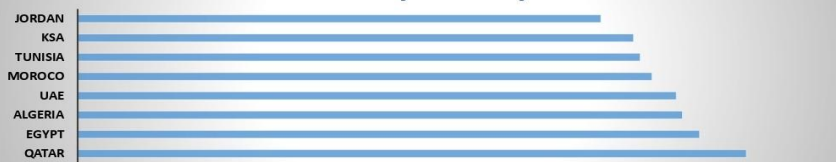
Sales By Category



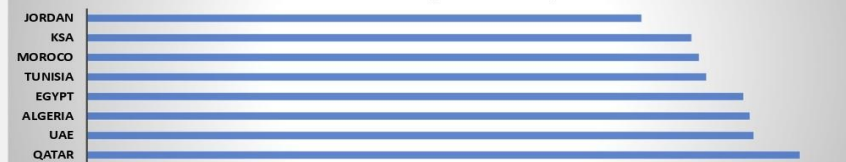
Profit By Category



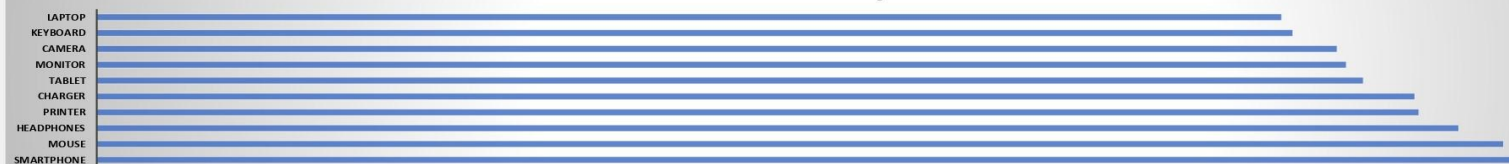
Sales By Country



Profit By Country



Sales By Product



Dashboard Features

Feature	Description
Total Sales	Overall sales amount
Total Profit	Total profit generated
Total Orders	Number of orders
Average Order Value	Average revenue per order
Slicers	Filter by Country, Category, and Product
Interactive Charts	Automatically update based on filters

Key Insights

- ❑ Mobile products generated the highest sales.
- ❑ Qatar recorded the highest sales among all countries.
- ❑ Sales remained relatively stable throughout the year.
- ❑ Interactive filters allow users to explore sales performance by product, category, country, and date.

Skills Demonstrated

- Data Cleaning
- Data Validation
- Data Transformation
- Power Query
- Power Pivot
- Data Modeling
- Pivot Tables
- Dashboard Design
- Interactive Reporting
- Microsoft Excel